BRIEF RESEARCH COMMUNICATION

Firth's penalized logistic regression: A superior approach for analysis of data from India's National Mental Health Survey, 2016

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ABSTRACT

The National Mental Health Survey of India (NMHS) was a ground-breaking nationwide study that harnessed a uniform, standardized methodology blending quantitative and qualitative approaches. Covering data from 12 states across diverse regions, its mission was to gauge the prevalence of psychiatric disorders, bridge treatment gaps, explore service utilization, and gauge the socioeconomic repercussions of these conditions. This initiative provided pivotal insights into the intricate landscape of mental health in India. One of the analyses planned for NMHS data was to undertake a logistic regression analysis with an aim to unravel how various sociodemographic factors influence the presence or absence of specific psychiatric disorders. Within this pursuit, two substantial challenges loomed. The first pertained to data separation, a complication that could perturb parameter estimation. The second challenge stemmed from the existence of disorders with lower prevalence rates, which resulted in datasets of limited density, potentially undermining the statistical reliability of our analysis. In response to these data-driven hurdles, NMHS recognized the critical necessity for an alternative to conventional logistic regression, one that could adeptly navigate these complexities, ensuring robust and dependable insights from the collected data. Traditional logistic regression, a widely prevalent method for modeling binary outcomes, has its limitations, especially when faced with limited datasets and rare outcomes. Here, the problem of "complete separation" can lead to convergence failure in traditional logistic regression estimations, a conundrum frequently encountered when handling binary variables. Firth's penalized logistic regression emerges as a potent solution to these challenges, effectively mitigating analytical biases rooted in small sample sizes, rare events, and complete separation. This article endeavors to illuminate the superior efficacy of Firth's method in managing small datasets within scientific research and advocates for its more widespread application. We provide a succinct introduction to Firth's method, emphasizing its distinct advantages over alternative analytical approaches and underscoring its application to data from the NMHS 2015–2016, particularly for disorders with lower prevalence.

Key words: Firth, India, National Mental Health Survey, penalized logistic regression

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THE NATIONAL MENTAL HEALTH SURVEY (2015–2016) OF INDIA

The National Mental Health Survey (NMHS) of India, conducted in 2015–2016,[1] represents a significant endeavor to address the growing concern of mental, neurological, and substance use disorders in the context of non-communicable diseases, which are increasingly recognized as major contributors to morbidity and disability. Over the past five decades, research studies have delved into the prevalence, patterns, characteristics, and determinants of various mental disorders, albeit with limited exploration of care-related issues, service delivery aspects, and system challenges. This survey, commissioned by the Ministry of Health and Family Welfare and conducted by the National Institute of Mental Health and Neurosciences, sought to provide good quality, scientific, and reliable information to inform mental health policies and programs at the national and state levels.

The NMHS 2015–2016, spanning 12 Indian states, employed a rigorous multistage, stratified, random cluster sampling method, encompassing adults aged 18 and older. Qualitative and quantitative techniques were applied to analyze data from 39,532 individuals across 720 clusters, yielding a final sample of 34,802. Diagnoses were made using the Mini International Neuropsychiatric Interview 6.0.0 and classified according to the International Classification of Diseases, 10th revision. The study assessed the cross-sectional prevalence and disability of psychiatric disorders and collected sociodemographic, health, treatment, and care-related information via specially designed questionnaires. The Sheehan Disability Scale was used to assess disability, and the socioeconomic impact was measured through assessment instruments derived from the World Health Organization – Disability Assessment Schedule 2.0. The survey revealed a lifetime prevalence of "any mental morbidity" at 13.67% and a current prevalence of 10.56%, with psychoactive substance use, mood disorders, and stress-related disorders being the most common. Prevalence was higher among males, urban residents, and those with lower education and income. Notably, the treatment gap for mental morbidity was 84.5%. NMHS is the largest survey of mental morbidity in India, highlighting the urgent need for enhanced mental health planning, delivery, and evaluation in the country, given that nearly 150 million individuals are affected by mental morbidity.[2]

Traditional maximum likelihood logistic regression

Logistic regression is a common approach for calculating adjusted odds ratios. It is an effective, robust statistical approach in medical research, especially given its applications in measuring the strength and direction of association, controlling the influence of confounding factors, and forecasting outcomes.[3] These odds ratios allow scientists to draw inferences and applications to public health and policy. Typically, logistic models are fitted using maximum likelihood (ML) software, which yields reliable statistical results when the model is reasonably accurate and the sample size is sufficiently large, ensuring a minimum of 4–5 subjects per parameter at each outcome level.[4] However, ML estimation may become unreliable when dealing with small datasets, rare exposures or outcomes, or substantial underlying effects, especially when these issues occur together.

Traditional logistic regression is a widely adopted method for modeling binary outcomes. However, its applicability diminishes significantly as the size of the dataset shrinks. Small datasets are an inherent aspect of scientific research. In many scientific disciplines, researchers grapple with the limitations imposed by limited sample sizes. Whether studying the prevalence of rare diseases, assessing the impact of a new drug, or exploring infrequent occurrences in social sciences, the need to analyze binary outcome variables with few positive instances is common. In small datasets, logistic regression often yields unstable parameter estimates, biased coefficients, and unreliable *P* values. These issues stem from the reliance on ML estimation (MLE), a standard technique in logistic regression. MLE aims to maximize the likelihood function to estimate model parameters, but its effectiveness diminishes when data is sparse and events are rare.[5]

A key shortcoming of traditional logistic regression in small datasets is the propensity for parameter estimates to exhibit substantial bias. This bias can distort the interpretation of coefficients, leading to incorrect conclusions. Moreover, small datasets may produce wide confidence intervals and unreliable hypothesis tests, further diminishing the confidence researchers can place in their findings.

Another critical issue arises when dealing with rare events.[6] In scientific research, rare events often hold particular importance as they may represent critical phenomena. However, traditional logistic regression underestimates the probabilities associated with rare events. The model exhibits a bias toward zero, making it challenging to assess the likelihood of such occurrences accurately. This bias can have far-reaching consequences, impacting decision-making processes and policy recommendations.

The challenges in applying logistic regression to the analysis of India's National Mental Health Survey 2016 data

Ideally, a logistic regression analysis benefits from a case-control data collection design. However, in the case of the NMHS, the data were not gathered in a prospective case-control format. Instead, it comprised survey data enriched with comprehensive sociodemographic variables. The challenge lay in using this survey data in a case-control design, a vital step to facilitate logistic regression analysis

for comprehending how predictor variables affect the dependent variable. Although logically possible, this poses some challenges.

Smaller prevalence disorders

Although the sample size in NMHS was large, the prevalence of most disorders was less than 1%. These datasets are characterized by a substantial imbalance between the events (ones) and non-events (zeros), and this disparity presents unique statistical challenges. For example, the separation can almost fully discriminate the presence or absence of the outcome in question, based either on a single predictor variable or a linear combination of several predictor variables.[7]

Data with high separation or high monotone likelihood

One of the other primary challenges of working with the NMHS dataset for logistic regression analysis is the risk of complete separation. Complete separation occurs when predictor variables perfectly predict the outcome, resulting in distinct groups where the event occurs only in certain combinations of predictor values. This leads to non-convergence in traditional logistic regression models, rendering them impractical or less useful for data analysis in such cases.

A hypothetical example

Let us consider a hypothetical dataset related to urban living and generalized anxiety disorder (GAD) and explain why traditional ML logistic regression can be problematic for the analysis due to separation or low sample sizes [Table 1].

Suppose you are conducting a study to examine the relationship between urban living (1 = Urban, 0 = Non-Urban), Gender (1 = Male; 2 = Female), and the diagnosis of GAD (1 = GAD present, 0 = GAD absent) in a sample of individuals.

Table 1: A hypothetical dataset of Generalized Anxiety Disorder diagnosis and associated socio-demographic variables

Subject	Urban living (Predictor)	Gender	Generalized Anxiety Disorder (Outcome)
1	1	2	1
2	1	2	1
3	0	1	0
4	0	1	0
5	0	1	0
6	0	1	0
7	0	1	0
8	0	2	0
9	0	2	0
10	0	2	0
11	0	2	0
12	0	1	0

Urban Living" (1=Urban, 0=Non-Urban), Gender (1=Male; 2=Female), and the diagnosis of Generalized Anxiety Disorder (GAD) (1=GAD present, 0=GAD absent)

In this dataset of 12 individuals, two persons screened positive for GAD. Here, once we have a closer look at the data, we observe a clear separation issue. If a person is living in an urban area (Urban Living = 1), they almost always have GAD (GAD = 1). Conversely, if a person is living in a non-urban area (Urban Living = 0), they almost always do not have GAD (GAD = 0). In addition, both the individuals who suffered from GAD belonged to the female gender (Gender code = 2). This represents a perfect or near-perfect association between the predictors urban living and female gender and the chances of diagnosis with GAD. Such challenges happen when either the overall number of persons with GAD in a larger epidemiological survey is less or when the overall sample size of persons with GAD is less within the larger sample of data.

Traditional ML logistic regression assumes that the predictors are not perfectly correlated with the outcome variable. However, in cases of perfect association, it can yield unstable parameter estimates, infinite odds ratios, and unreliable standard errors, as exemplified in this instance. Nevertheless, the NMHS 2016 dataset contained a wealth of epidemiological data with multiple explanatory variables, making it valuable to explore the impact of these variables on psychiatric disorder diagnoses. This analysis is instrumental in elucidating the relationships between predictor variables which can provide policymaking decisions, offering insights into this crucial aspect of mental health research.

Appropriate statistical approach

The team of authors and researchers examined alternative approaches to traditional ML logistic regression to mitigate the challenges described above. While there were several alternative techniques, the team decided to employ Firth penalized logistic regression, which introduces a penalty term to mitigate these issues, making the parameter estimates more stable and the results more reliable. The detailed advantages of Firth's penalized logistic regression and its applications are described below.

Firth's penalized logistic regression

Firth's penalized logistic regression, introduced by David Firth, addresses the limitations of traditional logistic regression when working with small datasets, particularly those characterized by rare events and complete separation. This method represents a significant advancement in statistical modeling, offering a robust solution for analyzing binary outcome variables in challenging settings.

Originating from the work of David Firth, Firth regression initially aimed to rectify the small sample bias commonly observed in traditional ML logistic regression.[8] Subsequently, it gained recognition as a powerful tool in scenarios where complete separation within the data prevents traditional ML logistic regression estimates from converging.[9] Firth regression belongs to the broader category of penalized

likelihood regression techniques, alongside methods such as least absolute shrinkage and selection operator (LASSO) regression and adaptive LASSO. The fundamental principle underlying Firth's regression involves introducing a penalty term into the standard ML function used to generate parameter estimates and standard errors for a logistic regression model. Notably, as the sample size approaches an infinite number of observations, this penalty term tends to converge toward zero, making Firth regression particularly well-suited for addressing small sample bias.[8]

The core principle underlying Firth's method is the incorporation of a penalization term into the likelihood function. This additional term serves to mitigate bias, stabilize parameter estimates, and ensure the convergence of the logistic regression model. The penalty term shrinks coefficient estimates toward zero, improving the accuracy and reliability of results.

Overall, Firth's penalized logistic regression leads to improved model fit, which is critical when working with small datasets. Firth's method enhances the goodness of fit by addressing bias and improving model stability. This results in a more accurate representation of the underlying relationships within the data. Specific advantages it offers are as follows: a) Bias reduction: Firth's method significantly reduces bias in parameter estimates, even in small samples. Introducing a penalty term that discourages extreme coefficient values counteracts the inherent bias observed in traditional logistic regression. This results in more accurate and stable estimates especially coefficient estimates[10] and overcomes issues related to model convergence.[11] b) Handling of rare events: Scientific research frequently deals with rare events that are of paramount importance. Firth's method excels at estimating probabilities for rare events as it overcomes the bias toward zero that plagues traditional logistic regression. Researchers can rely on Firth's method to obtain more precise and reliable estimates of event probabilities.[12]

The versatility of Firth's penalized logistic regression extends to various scientific domains with public health implications, especially when studying rare health outcomes or adverse events. Firth's method allows researchers to assess the likelihood of such events accurately, aiding in the development of preventive measures and healthcare interventions. Researchers across disciplines have embraced this method as a powerful tool for tackling the challenges associated with small datasets. Some notable applications are as follows: a) Medical research: Small sample sizes are common in medical research, particularly when investigating rare diseases or assessing the efficacy of novel treatments. Firth's method ensures that researchers can obtain accurate estimates of event probabilities, even when dealing with limited patient data.[9,13] b) Environmental science: Environmental studies often involve analyzing infrequent events, such as extreme weather phenomena or rare ecological occurrences. Firth's method enables researchers to model these events effectively, providing valuable insights into environmental processes. c) Social sciences: Rare events frequently occur in the social sciences, from political coups to economic crises.[14] Researchers in these fields benefit from Firth's method's ability to produce reliable estimates, even when working with sparse data.

Alternative approaches to Firth penalized logistic regression

There are several other approaches to small datasets and datasets with wide separation.

Non-penalized approaches

These include exact logistic regression (useful when there is complete separation as it directly calculates the exact *P* values for logistic regression coefficients; can be modified to conditional logistic regression if there are matched pairs or strata), Bayesian logistic regression (the additional advantage is that this allows for the incorporation of prior information about the relationships between predictors and outcomes, helping to stabilize parameter estimates), and bootstrapped logistic regression (involves repeatedly resampling the replacements to create multiple subsamples; this approach repeatedly performs logistic regression on each of these subsamples, thus computing more reliable estimates and confidence intervals).

Penalized approaches

These include Lasso and Ridge regularizations, which basically add a penalization to the likelihood function, thus reducing overfitting and multicollinearity. Lasso regularization can even perform variable selection, especially when some predictor variables are not important or relevant.

Alternative modeling techniques

Alternative modeling such as decision trees, random forests, and support vector machines can handle complex relationships between variables despite small sample sizes.

Application of Firth's penalized logistic regression to National Mental Health Survey of India 2016

Analyzing the rich dataset from this survey is a complex task, especially in disorders with lower prevalence. Firth penalized logistic regression offers a viable method to address the unique challenges and nuances inherent to psychiatric research in India, as well as in analyzing the predictors of a particular disorder.

Psychiatric research often faces the issue of separation, where a predictor perfectly separates the outcome variable, rendering standard logistic regression models unreliable. In the context of the NMHS, where the relationships between risk factors and psychiatric disorders are undoubtedly intricate, Firth penalized logistic regression is a valuable

solution. This technique introduces a penalization term into the likelihood function, effectively alleviating the separation problem. By shrinking the ML estimates toward a specific value, the method ensures that the model remains stable and reliable, even in cases where traditional logistic regression would fail. This is particularly relevant in a dataset with a broad range of sociodemographic variables and the complex interplay of factors influencing mental health outcomes in the diverse Indian population.

Moreover, small sample sizes can be a concern in psychiatric research, especially when dealing with the regional and sociocultural diversity of the NMHS. Firth penalized logistic regression is well-suited to this scenario. It helps prevent overfitting by introducing a penalty term, which effectively controls the complexity of the model. Overfit models, which perform well on the data they were trained on but generalize poorly to new data, can be a significant issue when there are many predictors and limited observations. By imposing a penalty that discourages overly complex models, Firth penalized logistic regression maintains the model's ability to generalize, ensuring that the results are robust and not driven by noise in the data.

In addition, multicollinearity is often a concern when analyzing such an extensive and complex dataset. Multicollinearity occurs when predictors in a model are highly correlated, which can lead to unstable and unreliable coefficient estimates. Multicollinearity is a real threat given the numerous sociodemographic variables collected in the NMHS. Firth penalized logistic regression addresses this problem by regularizing the coefficient estimates, reducing the impact of correlated predictors, and making the model results more interpretable.

Furthermore, Firth penalized logistic regression is particularly adept at handling datasets with imbalanced outcomes, a common issue in psychiatric research. In studies of mental health disorders, the prevalence of certain conditions may be low, leading to imbalanced data. Traditional logistic regression models can produce biased results when applied to such datasets. Firth penalized logistic regression, by introducing a correction factor into the likelihood function, which can mitigate this bias, ensuring that the results accurately reflect the true relationships between predictors and outcomes, even in the presence of imbalanced data.

Practical considerations

In the context of analyzing the data from the NMHS, where the prevalence of mental disorders has been carefully documented, a strategic approach to statistical modeling is imperative. When dealing with disorders exhibiting a lifetime prevalence of > 1, such as alcohol use disorder, depressive disorders, and tobacco use disorders, logistic regression with the ML function may be still considered as a suitable method. The choice between traditional logistic regression and Firth penalized logistic regression is informed by the prevalence of the psychiatric disorders under investigation, considering the complexity of the predictor variables in our study. When dealing with disorders showing a prevalence of 1% or higher, and with our dataset size surpassing 1% of the total population, which equates to more than 350 cases within our study of 34,802 individuals, traditional logistic regression offers a stable foundation for analysis. For psychiatric disorders with a prevalence of less than 1%, as encountered in subcategories of these predictor variables, the study authors have opted for Firth penalized logistic regression. This decision is especially pertinent given the extensive array of predictor variables, including gender, education, employment status, marital status, urbanicity of residence, age groups, and income, with more than 15 subcategories across them. This approach effectively addresses the challenges posed by sparse data, ensuring more reliable insights into these less frequent yet significant mental health conditions.

By employing ML estimation, we can accurately model the relationships between these more prevalent disorders and various covariates, providing valuable insights for crafting effective public health interventions and policies. Conversely, for disorders with a lifetime prevalence of < 1, such as schizophrenia disorder, post-traumatic stress disorder, and social anxiety disorder, Firth penalized logistic regression emerges as a valuable tool. These disorders are characterized by lower prevalence, which can lead to statistical challenges such as separation or quasi-complete separation in the data. Firth penalized logistic regression helps address these issues by mitigating bias and improving parameter estimation, even when dealing with sparse data. For disorders in this category, Firth penalized logistic regression ensures the accuracy and stability of odds ratios, which are crucial for understanding the risk factors and associations associated with less frequent mental health conditions. This tailored approach, adapting to the specific prevalence of the disorders in question, demonstrates the importance of selecting the right statistical method to draw meaningful insights from the survey data.

Implementing Firth's penalized logistic regression in scientific research is facilitated by the availability of software tools and packages. Statistical software environments such as R[15] offer specialized functions for fitting Firth's models, making them accessible to researchers across disciplines. These software tools streamline the implementation of Firth's method and provide researchers with user-friendly interfaces for conducting analyses.

CONCLUSION

In the realm of scientific research, where the analysis of small datasets is a recurrent challenge, Firth's penalized

logistic regression emerges as a good choice for discerning statisticians and researchers. As scientific exploration increasingly delves into the intricacies of rare events, or data with separation, Firth's penalized logistic regression remains a useful statistical tool, enabling researchers to extract valuable insights. This preference stems from its ability to address the pervasive issue of bias, adeptly handle rare events, ensure model convergence, and consistently deliver dependable and precise results. Particularly in the context of the NMHS data analysis of disorders with lesser prevalence, characterized by its extensive dataset complexities, Firth's penalized logistic regression stands out as the superior choice to traditional ML logistic regression.

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Conflicts of interest

There are no conflicts of interest.

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